This assignment will not be graded and is only for practice.

Level 1

Consider an ordered basis $\gamma = \{b_1, b_2, b_3\}$ of the inner product space $\mathbb{R}^3$ with respect to the dot product, where $b_1 = (1, 2, 3), b_2 = (1, 1, 1)$ and $b_3 = (0, 2, 1)$. Let $\beta = \{a_1, a_2, a_3\}$ be the orthonormal basis which is obtained using the Gram-Schmidt process from the basis γ. Use this information to answer the questions 1 and 2.

1) The vector a_2 is **1 point**

- $(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}})$
- $(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{-2}{\sqrt{21}})$
- $(\frac{-4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}})$
- $(\frac{4}{\sqrt{21}}, \frac{-1}{\sqrt{21}}, \frac{2}{\sqrt{21}})$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{-2}{\sqrt{21}})$

2) The vector a_3 is **1 point**

- $\frac{1}{\sqrt{6}}(-1, 2, -1)$
- $\frac{1}{\sqrt{6}}(-1, -2, -1)$
- $\frac{1}{\sqrt{6}}(1, 2, -1)$
- $\frac{1}{\sqrt{6}}(1, 2, 1)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\frac{1}{\sqrt{6}}(-1, 2, -1)$

Level 2

3) Let $v_1 = (1, 0, 1, 1)$ and $v_2 = (0, 1, 1, 1)$ be the vectors from the inner product space $\mathbb{R}^4$ with respect to the dot product. If $v_3 = v_2 + av_1$ where $a \in \mathbb{R}$ and v_1, v_3 are orthogonal, then **1 point**

- $a = -2/3$
- $a = 2/3$
- $a = 1/3$
- $a = -1/3$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$a = -2/3$

4) Let W be a subspace of the inner product space $\mathbb{R}^4$ with respect to dot product and $\{v_1, v_2, v_3\}$ be an ordered basis of W, where $v_1 = (1, 1, 1, 1), v_2 = (1, 1, 1, 0), v_3 = (1, 1, 0, 0)$. Which of these is the orthonormal basis of W obtained using the Gram-Schmidt process? **1 point**

- $\{\frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(1, 1, -2, 0)\}$
- $\{\frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(1, 1, 2, 0)\}$
- $\{\frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(3, 1, -2, 0)\}$
- $\{\frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(3, 1, 2, -2)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{\frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(1, 1, -2, 0)\}$

5) **1 point**

Let $a = (\frac{2}{3}, \frac{2}{3}, \frac{1}{3})$ be a vector from the inner product space $\mathbb{R}^3$ with respect to dot product and $W = \{(x, y, z) \in \mathbb{R}^3 \mid \langle (x, y, z), (\frac{2}{3}, \frac{2}{3}, \frac{1}{3}) \rangle = 0\}$ be a subspace of $\mathbb{R}^3$. Then which of the following is (are) a basis of W?

- [ ] $\{(1, 0, 2), (0, 1, 2)\}$
- [ ] $\{(1, 0, -2), (0, 1, -2)\}$
- [ ] $\{(-1, 0, 2), (0, -1, 2)\}$
- [ ] $\{(1, 0, 2), (0, 1, -2)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 0, -2), (0, 1, -2)\}$
$\{(-1, 0, 2), (0, -1, 2)\}$

Your score is: 0/5